Section 3.7 — Radical Equations

Math 103: College Algebra

Solutions to Radical Equations

Decide whether each value is a solution to the given equation.

1. $1 + \sqrt{x} = \sqrt{2x+1}$; $x = 4$
2. $\sqrt[3]{3x-4} = \sqrt[3]{5x+2}$; $x = -3$
3. $\sqrt{x^2 - 3x + 5} = \sqrt{x^2 - 7x + 9}$; $x = 2$
4. $\sqrt[3]{2x+4} + \sqrt{1-5x} = \sqrt[4]{x^2 - 3x + 6}$; $x = -2$

Equations with an Isolated Radical

Solve. Be sure to eliminate any extraneous solutions. If there is no solution, write "no solution."

5. $\sqrt[3]{x} = -2$
6. $\sqrt{x+3} = 5$
7. $\sqrt[4]{x+7} = -1$
8. $\sqrt{x^2 - 4} = x + 2$
9. $\sqrt{21 + 4x} = 2x + 3$
10. $\sqrt[3]{y^2 + y + 7} = 3$

Equations with One Radical

Solve. Be sure to eliminate any extraneous solutions. If there is no solution, write "no solution."

11. $\sqrt{y-6} + 2 = 9$
12. $\sqrt[3]{n+3} - 2 = -4$
13. $2\sqrt[4]{m+3} + 4 = 2$
14. $x - \sqrt{x} = 6$
15. $\sqrt{6v+1} - 1 = v$
16. $\sqrt{x^2+4} - x = 4$
17. $x - \sqrt[3]{x^3 - 7} = 1$
18. $\sqrt[3]{5x^2 + 28x + 31} - x = 3$

Equations with Two Radicals

Solve. Be sure to eliminate any extraneous solutions. If there is no solution, write "no solution."

19. $\sqrt{m+2} = \sqrt{2m-3}$
20. $2\sqrt[4]{3h-4} - 3\sqrt[4]{h+4} = 0$
21. $2\sqrt{x+6} - \sqrt{4x+24} = 0$
22. $\sqrt{4x-5} = \sqrt{6x+5}$
23. $\sqrt{-5x+56} = \sqrt{-x+44} - 2$
24. $\sqrt{24w-95} - 2\sqrt{5w-20} = 1$
25. $\sqrt{-5d+41} = 3 + \sqrt{-11d+56}$
26. $3\sqrt{-5w+11} = \sqrt{-33w+67} - 2$

Inverse of a Radical Function

Give the inverse of each radical function. If it is an even root, be sure to include any required restrictions on the domain.

27. $f(x) = \sqrt[3]{x}$
28. $g(x) = \sqrt[6]{x}$
29. $h(x) = -\sqrt{x}$
30. $k(x) = \sqrt[5]{x-3} + 2$
31. $s(x) = \frac{1}{2}\sqrt{2x-5} - 7$
32. $t(x) = -2\sqrt[3]{7-x} + 4$
33. $v(x) = \sqrt[42]{3x-5}$

Inverses of Power Functions

Give the inverse of each power function. Note the domain restrictions.

34. $f(x) = x^3$
35. $g(x) = x^6,\ x \geq 0$
36. $h(x) = x^4,\ x \leq 0$
37. $k(x) = x^3 - 5$
38. $s(x) = (x-2)^7$
39. $t(x) = (4-x)^2,\ 4 - x \geq 0$